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# Micro and mesoscale combustion

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**Abstract**

A review of research and development on micro and mesoscale combustion is presented, with an emphasis on fundamental understandings achieved in the field during the last decade. Due to its small scale nature, increasing effects of flame–wall interaction and molecular diffusion are the characteristic features of micro and mesoscale combustion. After brief review of device developments, overview of fundamentals in micro and mesoscale combustion as well as possible future directions is presented.
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*Keywords:* Microscale combustion; Mesoscale combustion; Microcombustion

## 1. Introduction

With the recognitions of broad potential applications not only for electrical power but for heat and mechanical power sources, combustion at small scales (micro and mesoscales) is collecting growing attentions these days [1–4]. Possible applications are sensors, actuators, portable electric devices, robots, rovers, unmanned air vehicles, thrusters, industrial heating devices, and furthermore, heat and mechanical backup power sources for air-conditioning equipments in hybrid vehicles and freight transportations as well. Needless to say, the concept is based on the nearly two-order higher energy densities of hydrocarbon fuels than the existing modern batteries.

Along with the demands for further developments of small scale combustion devices, fundamentals on micro and mesoscale combustion also collect increasing interests through the technical challenge to overcome quenching issues due to the large surface-to-volume ratio of small scale devices. Thermal and chemical stability managements are required to establish stable combustion in micro and mesoscale devices. Thermal stability managements by the heat recirculation displayed the several novel phenomena such as flame bifurcation, weak flame, extinction and re-ignition instabilities, spinning flames and pattern formations. From the aspect of flame–wall chemical coupling, new concept catalytic combustion and non-equilibrium effects at wall on combustion are started to be explored. Understanding on oxygen absorption led to extremely low-temperature operation of microscale catalytic combustion. In this paper, recent applications and fundamentals of micro and mesoscale combustion are reviewed and summarized. Future directions and expectations will be also discussed.

## 2. Scaling for microscale combustion

In small scale combustion systems, surface-to-volume ($S/V$) ratio is large due to the small characteristic length scale. The large $S/V$ ratio leads to the severer heat-loss effect on small scale

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combustion systems. Hence, clear understanding on flame extinction, which is governed by the ratio of heat-loss to the heat generation described in theory of non-adiabatic flames [5,6], is important. Discussion on quenching diameter can be made by assessing the heat loss from a flame with finite diameter. It is essentially equivalent to the discussion based on Peclet number, in which the ratio of quenching diameter to the flame thickness is considered.

Discussion on flame extinction by Peclet number, defined as the ratio of longitudinal to radial heat transfers, is also of use. It can be widely applied to flame extinction by replacing loss mechanisms. Simple comparison between residence and chemical time scales by Damköhler number is also useful for blow-off type extinction.

Reynolds number is generally small due to the small characteristic length scale in small scale system. Therefore, flow condition in microscale system often remains in laminar, while it sometimes reaches to the transition regime in mesoscale system. Since turbulent mixing is least expected, molecular diffusion is primary mechanism for mixing. Thus, molecular diffusion may control overall characteristics or performance of microscale systems with low velocity forced flow or non-premixed flames.

To overcome the effect of large heat loss on small flames, stability management is required. One of the effective stability managements is heat recirculation through channel wall. To understand the characteristics of heat recirculation process, Biot number, which is defined as the product of heat transfer coefficient and characteristic length divided by thermal conductivity of the solid, is of use. It is denoted as thermally thin wall if $B < 0.1$, where the heat resistance in the solid phase is small and surface heat resistance is predominant. For transient process, thermal and mass diffusion characterized by Fourier number, defined by the ratio of the product of thermal or mass diffusivity and characteristics time scale divided by the square of the characteristic length scale, is also important. Accordingly, characteristics of heat recirculation process for a stationary flame in a channel and those for propagating flame are quite different.

Unequal thermal and mass diffusivities often lead to manifestations of the diffusive-thermal instabilities. In microscale combustion with heat generation, heat transfer through wall is combined with gas-phase heat transfer, resulting in modified thermal and mass transfer balance. Thus, effective Lewis number should be estimated for describing the onsets of instabilities based on the diffusive-thermal instability.

If the length scale of the system is extremely small (or it is under the reduced pressure), Knudsen number, defined as the ratio of mean free path to the characteristic length scale, becomes large and the continuum assumption of fluid mechanics is no longer a good approximation. Although current microscale combustion does not often address this issue, some attempts have been made in this direction by paying attention on the concentrations and temperature discontinuities at wall.

In this paper, the term "microscale combustion" is used for combustion which involves characteristic length scales closely related to "quenching distance," particularly when it plays critical role on the addressed phenomena. Therefore, topics with length scales not only smaller than $1 \times 10^{-3}$ m physical-scales are dealt in this paper. The term "mesoscale combustion" will also be used for the cases with characteristic length scales obviously larger than 1 mm but remains characteristic features of microscale combustion. It is often used to highlight specific features of microscale combustion using scale-up model approach.

## 3. Application of microscale combustion

### 3.1. Microscale combustion for mechanical power

#### 3.1.1. Microscale gas-turbines and internal combustion engines

Based on the high energy density of hydrocarbon liquid fuels, scale-down approach of existing rotating machinery and internal combustion engines for portable power sources have been conducted.

MEMS-based silicon gas-turbine engines were developed at MIT [7–10]. The micro gas-turbine is a 10 mm diameter by 3 mm thick, toward electrical power output up to 20 W at rotor speed of $2.4 \times 10^6$ rpm, consuming 7 g/h of $H_2$ fuel. Hydrogen-fueled micro combustor produced exit gas temperature in excess of 1800 K by a micro combustor having 66 $mm^3$ chamber [11]. Stable hydrocarbon combustions were also demonstrated by a micro combustor having 195 $mm^3$ chamber [9] and six-wafer catalytic micro combustor [12].

Tanaka, Esashi and co-workers at Tohoku, collaborating with IHI, demonstrated that the world's smallest-class gas-turbine which operates Brayton cycle [13]. It has a compressor of 16 mm diameter, a turbine of 17.4 mm diameter, an annular combustor and a dummy electromagnetic generator. The impellers are connected by one piece of Inconel shaft of 8 mm diameter as shown in Fig. 1 right. Brayton cycle was established when pressure at the compressor inlet became negative (minus a few kPa below the atmospheric pressure) at the rotation speed of 360,000 rpm and the combustor temperature of 800–900 °C. Hydrogen was used for the demonstration.

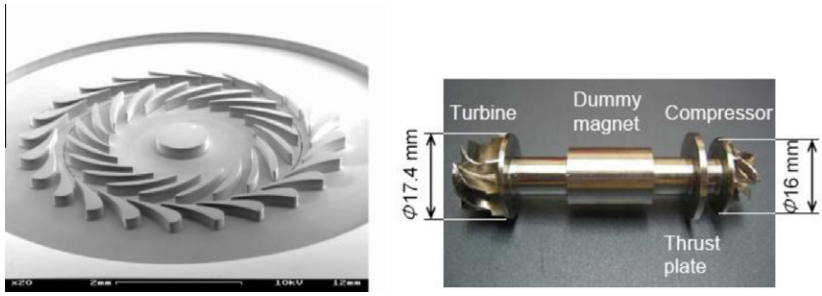


Fig. 1. Left: MEMS-based gas-turbine engines developed at MIT. Right: Rotor for miniature gas-turbine developed at Tohoku with IHI.

Meso- and microscale rotary engines by electro-discharge machining (EDM) and MEMS-based fabrications were developed at UCB [14–16]. Hydrogen-fueled, mesoscale rotary engine with 10-mm rotor produced 4 W at 9300 rpm. Meanwhile, research was centered on the fabrication of the individual engine components in the case of MEMS-based micro rotary engines having 2.4 and 1.0 mm rotors,. Goal of the development was set at 90 mW power output from 2.4 mm engines (see Fig. 2).

A mesoscale single-shot free-piston rapid compression machine was constructed for examining the feasibility of miniature HCCI engine at Honeywell in collaboration with University of Minnesota [17,18]. Figure 3 shows piston and cylinder assembly. The cylinder is Pyrex tube with nominal inside diameter of 3.0 mm. *n*-Heptane/air mixtures with the equivalence ratio at 0.69 and 0.25 were successfully ignited with piston stroke at 57 mm. HCCI combustion was possible in a combustion volume with 3 mm diameter and 0.3 mm long. A single-shot zero-dimensional numerical model reproduced piston position and velocity measurements which predicted *P–V* diagram with the effect of the leakage on the efficiency of mini HCCI engines for various fuels (*n*-heptane and propane).

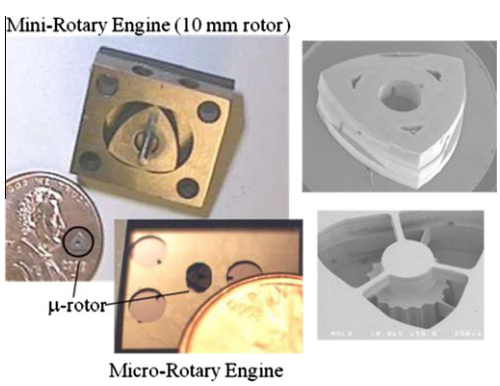


Fig. 2. Meso/microscale rotary engines developed at UCB.

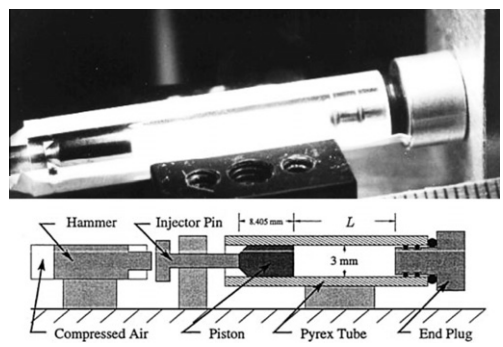


Fig. 3. The mesoscale free-piston "knock" engine using homogeneous charge compression ignition (HCCI). Upper: picture of cylinder and piston assembly. Lower: scheme of single-shot experimental apparatus.

A direct power generation by a ferromagnetic free-piston and cylinder, surrounded by the permanent magnetic MEMS arrays were developed at Georgia Tech [19]. From the 43-mm stroke and 13,400-mm$^3$ displacement, it produced 12 W of the electrical power while efficiency was not high due to the severe sealing issue. For light-weight power generation, a miniature internal combustion engine (MICE) generator, a two-stroke liner-motion free-piston engine with a linear electric alternator was developed at Aerodyne Research [20].

An engine with swinging free-piston termed "Mesoscale internal combustion swing engine (MICSE)" with four combustion chambers formed with swinging free-piston and combustor-body, was developed at University of Michigan (Fig. 4) [21,22]. The engine was designed to operate in a four-stroke Otto cycle and energy conversion is made by an inductive alternator connected to the shaft of the swinging piston.

### 3.1.2. Micro thrusters

Micro and mesoscale thrusters with various fabrication techniques have been developed for space applications such as microsatellite primary

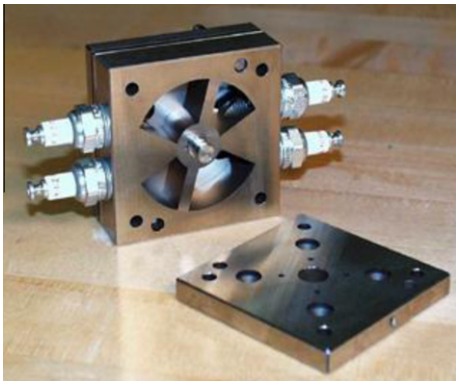

Fig. 4. Mesoscale internal combustion swing engine (MICSE) with four combustion chambers.

propulsion and attitude control [23–27]. Digital propulsion micro-thruster, which has a three-layer sandwich of silicon and glass layers, consists of micro-resistors, thrust chambers and rupture-diaphragms, was developed [23]. The sandwich contains an array of small plenums each sealed with a diaphragm on one side. Arrays of the thrusters were integrated in a 24-pin ceramic DIP (Fig. 5, upper left). Using lead styphnate as a propellant, a unit thruster produced $10^{-4}$ N s of impulse and about 100 W of power. A high pressure bipropellant micro-thruster by fusion bonding of stacked six silicon wafers (Fig. 5, upper right) was developed at MIT [27]. It produces 1 N of thrust from a mass of 1.2 g chamber and specific impulse is approaching 300 s.

Micro and mesoscale vortex flow combustors using gaseous non-premixed combustion for use in small scale power and propulsion systems are developed [28]. Series of combustors, ranging in size from 10.6 to 124 $mm^3$ which correspond combustor diameters of 2.4–6.4 mm, were fabricated

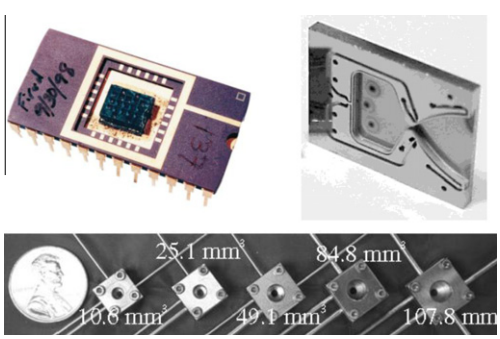


Fig. 5. Upper left: Digital micro-thruster developed at TRW. Upper right: High-pressure bipropellant micro-thruster by stacked silicon wafers developed at MIT. Bottom: Micro/mesoscale vortex flow combustors developed at Penn State.

by EDM for scaling studies (Fig. 5, bottom). Chemical efficiencies were exceeding 97% for hydrogen/air and 85% for methane/oxygen-enriched air. Numerical analysis identified flame stabilization by a nearly quiescent region of hot combustion products in the combustor core which was created by flow recirculation. Liquid-propellant chemical micro-thrusters were also developed from aluminum oxide using ceramic stereo-lithography [25].

Low-temperature co-fired ceramic (LTCC) technology [29] was adopted for fabricating microscale thrusters [26,30,31], since it enables to construct multi-layered three-dimensional structure with highly conductive materials such as silver, copper and gold, for realizing ceramic package with control wiring circuit.

LTCC micro-thrusters were developed using gunpowder-based solid propellant [31]. It produced $3.81 \times 10^{-5} - 1.27 \times 10^{-4}$ N s of total impulse and 5.55–14.41 s of specific impulse. A monopropellant micro-thruster ignited by electrolysis based on LTCC technology was developed [32]. The volume of the combustion chamber was 0.82 $mm^3$. Screen printed silver electrodes were positioned on the top and bottom surfaces of the combustion chamber. DC voltage potential applied across the electrodes initiates decomposition of hydroxylammonium nitrate (HAN) based liquid monopropellants. A thrust output of 150 mN was obtained at voltage input of 45 V.

Furthermore, various types of micro and mesoscale thrusters, mega-pixel thruster arrays [24], a two-staged mesoscale combustor, which consists of a microscale catalytic reactor and a mesoscale quartz main combustor [33] and a mesoscale chemical thruster with catalytic decomposition of hydrogen peroxide ($H_2O_2$) [34], were developed.

### 3.2. Microscale combustion for heat source

Toward power generation, two-dimensional Swiss-roll (Fig. 6a) and three-dimensional toroidal (Fig. 6b and c) burners were developed at USC [35]. To maximize heat recirculation and minimize external heat loss, a toroidal three-dimensional Swiss-roll burner was proposed [35] based on the two-dimensional Swiss-roll burners, of which original two-dimensional burner concept suggested by Weinberg [36,37]. For power generation, a number of thermoelectric devices are embedded in all the diving walls of these burners [38] by developing a special fabrication technique. Despite the great efforts, fabrication of thermoelectric devices with high figure-of-merit in such complicated multi-dimensional configuration remains a difficult task. Theoretical investigation for efficient energy conversion by thermoelectric devices with a heat recycling burner is examined

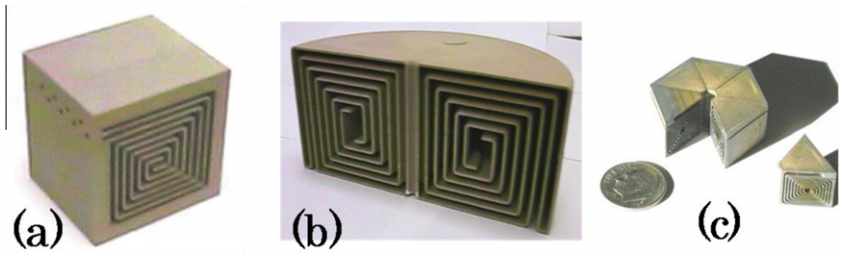


Fig. 6. Burner prototypes fabricated at USC. (a) Two-dimensional Swiss-roll burner, (b) macro and (c) mesoscale toroidal three-dimensional burners. Both macro and mesoscale versions attained self-stabilized combustion.

[39]. Two-dimensional Swiss-roll burner was successfully applied to hydrocarbon-fueled, thermally self-sustained, and single-chambered micro SOFC by the same group [40].

A different-type ceramic Swiss-roll microreactor (12.5 × 12.5 × 5.0 mm) coupled with thermoelectric devices was also developed with modified stereo-lithography [41].

By taking the advantage of high energy density of liquid hydrocarbon fuels, mesoscale catalytic combustors using electrosprayed liquid hydrocarbons, which will be coupled with energy conversion modules, were developed [42,43]. Electrospray atomization relies on electric charging of liquid fuel with sufficient electric conductivity. Under the electric field, leading edge of the liquid jet at the outlet of the nozzle breaks up toward the downstream and atomized into a spray of fine, charged droplets (Fig. 7) [43]. The electrospray enables quasi-monodispersity of the droplets within the order of the 10% deviation from the relative standard droplet size. Coulombic repulsion of the charged droplets prevent droplet coalescence resulting in mixing enhancements.

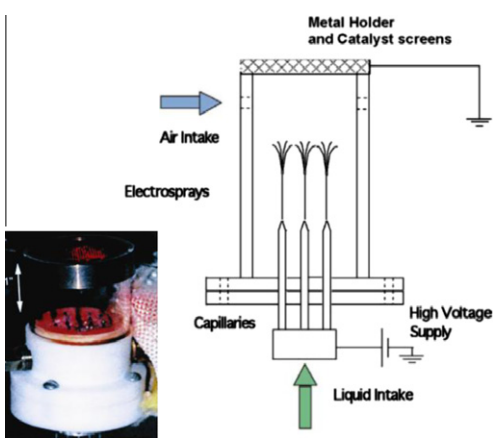


Fig. 7. A catalytic mesoscale burner (left) and electrospray injection system (right) with catalytic screens which act as a uniform and stable high temperature source for energy conversion modules.

JP-8 at flow rates of 10 ml/h which corresponds to 100 W of thermal power (~6.7 MW/m$^3$ of volumetric power density at 15 cc of combustor volume) was achieved. For higher energy density, miniaturization through the combination of microfabricated multiplexed electrospray system was conducted [44]. The same order of thermal power at one or more order smaller combustor in volume was established. The corresponding total energy density 270 MW/m$^3$ is comparable to those of conventional macroscale gas-turbine engines operated at elevated pressures. JP-8 electrosprayed catalytic combustor was combined with a free-piston Stirling engine. It recorded much higher efficiency though the system is in larger scale [45].

Disk-shaped Swiss-roll microcombustors for general purpose heat sources were developed at Tohoku [46,47] (Fig. 8). Several prototype combustors with outer diameters of 64, 45, 26 and 20 mm with channel widths smaller than the ordinary quenching distance were fabricated. By varying the ratio of channel width and height, the ratio of the heat emission from top and bottom panels to the outside, to the incoming mixture can be designed. Larger prototypes (64 and 45 mm) exhibited wide turn-down ratio at the averaged inlet mixture velocity from 0.4 to 8.0 m/s. Blow-off limit was not observed at large combustors due to the high temperature limit of the employed material (SUS304). Established surface temperature range was from 400 to 1173 K.

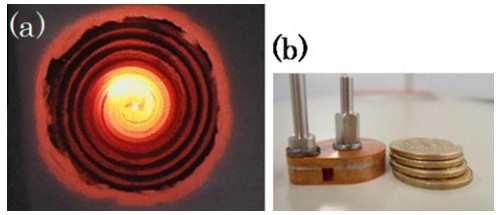


Fig. 8. Disk-shaped Swiss-roll combustors for general purpose heat sources. (a) A 64-mm Swiss-roll combustor in operation; a top panel replaced by a quartz plate for visualization. (b) Coin-size combustor (26 mm, o.d.).

Temperature controllability of the heater panel was within 1 K when simple feed-back control was used. Total thermal efficiency as a heater was 60–85%, which were 1.5–2 times larger than those of conventional resistive heaters. Simple analytical model was constructed to examine the effect of design parameters on combustion characteristics. Based on the model, the possible smallest self-sustaining Swiss-roll combustors with 26 and 20 mm were fabricated by EDM and tested. The model predictions were excellent and narrow stability windows with both lower and blow-off limits were observed.

A small combustor having a stabilized annular flat flame of inner and outer diameter of 5 and 10.5 mm, which would fit the MIT micro gas-turbine was fabricated by Yuasa et al. [48]. Flame stabilities and effect of design parameters on extinction limits in the developed combustor were examined for hydrogen/air and methane/air mixtures.

"Mesoscale liquid-fuel-film combustor" was developed at UCI [49,50]. The fuel is injected as a liquid film on the inner surface of cylindrical combustor and swirling air flow provides effective fuel vaporization. It enables reduction of heat losses by redirecting heat towards film vaporization and cools the combustor walls. Due to the large $S/V$ ratio in small scale, large liquid surface-area is expected for vaporization. It was shown that internally stabilized flame was attained for liquid-fueled flames but not for gas-fueled flames. Figure 9 shows the first prototype Pyrex-glass burner (left) and temperature profiles at 2 mm above the metal burner exit from CARS measurements for heptane flame at overall $\phi = 1.69$ (right). The latter shows that measured temperatures are below the stoichiometric adiabatic flame temperature, but generally above the fully premixed adiabatic flame temperature. This suggests that the flame is partially premixed, with near stoichiometric conditions along the central axis, and very rich regions near the walls. For the observation of the film and flame behavior, a sapphire/metal burner was also developed.

Ishizuka and co-workers [51] also examined flame propagation in a vortex flow within a mesoscale tube and confirmed the effect of maximum tangential velocity on the onset of the vortex bursting.

### 3.3. Current achievements in micro power generation

Demonstrations of catalytic or non-catalytic combustion toward power generation with thermoelectric [52–54] and TPV devices [55–57] were also conducted.

Catalytic microreactors with high-precision tape-casting method [58] applied for butane accomplished stable operation. Thermal energy density around $2–8 \times 10^8$ W/m$^3$ [59,60] are achieved for higher temperature application (1073–1173 K) such as TPV devices. Integrated power generation system of hydrogen-fueled catalytic combustion on $TiO_2$ supported Pt with separated TE elements ($Bi_2Te_3$–$Sb_2Te_3$ for p-type and $Bi_2Te_3$–$Bi_2Se_3$ for n-type elements) recorded 184 mW output from 6.6 W chemical energy input which corresponds to the total efficiency of 2.8% [52]. Methanol-fueled catalytic microreactor system of Pt on anodized aluminum with commercial thermoelectric devices successfully produces 0.65 W electricity output which corresponds to the total efficiency 1.1% [61]. The combination of catalytic combustor and TE devices attained an overall efficiency of converting chemical energy to electricity of ~0.8% [53].

A micro TPV power generator system, consists of hydrogen-fueled cylindrical SiC microcombustor (3 mm in diameter and 16 mm in length), a nine-layer dielectric filter, and a GaSb (gallium antimony) photovoltaic cell array, generated 1.02 W electricity output from 140.1 W chemical

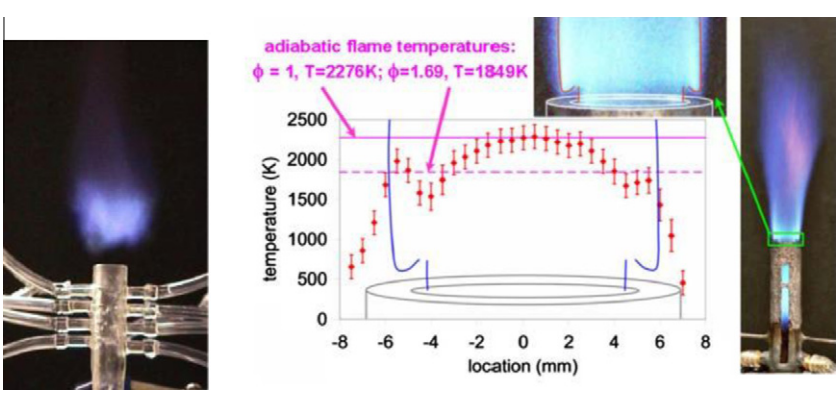


Fig. 9. Mesoscale liquid-fuel-film glass burner (left) and temperature profile at 2 mm above burner exit from CARS measurements for heptane flame of overall $\phi = 1.69$ (right).

energy input which corresponds to 0.73% total efficiency [56]. Combination of reactor with fuel reformer for hydrogen production is also attempted [62,63]. Characteristics of co-current and counter-current flow reactor reformer for $NH_3$ reforming on Ru are numerically examined [62]. Effect of heat recirculation on the performance of catalytic microcombustor is also investigated using catalytic reactor with heat recirculating channel [64]. CFD results showed that the excess enthalpy effect is substantial only in the limit of low-conductivity walls, where there is insufficient preheating of the reactants. Catalytic reactor as a heat source for micro endothermic reactor is fabricated using catalyst bed made from Isolite B5 coated with Pt by the incipient wetness method [65].

Recently, a narrow ceramics tube of i.d. 0.8 mm with porous catalytic layer inside the tube was used as combustor and it was combined with commercial Bi–Te TE modules. Total conversion efficiency of 3.0% was achieved for methane [54].

## 4. Fundamentals of microscale combustion

### 4.1. Flame quenching and stability managements

For the discussion on flame quenching, one-dimensional planar premixed flame with volumetric heat loss is considered [5,6]. Solution for non-dimensional form of equation is given as $m^2 = \exp(-l/m^2)$, where $m$ and $l$ are non-dimensional flame speed and heat loss parameter. For instance, fundamental flammability limit can be given by providing radiative heat loss from the burnt gas to the ambient as a loss mechanism. Here, quenching diameter is addressed by considering convective loss from a flame with a finite diameter [4,6]. Heat balance can be given as:

$$H = \frac{(\text{Heat loss})}{(\text{Heat generation})} = \frac{c_p(T_f - T_u)\omega(\pi d^2/4)\delta_f}{\pi d \delta_f h(T_f - T_u)} = 4Nu\delta_f^2/d^2.$$

Since extinction condition for the above solution is given as $m = 1/\sqrt{e}$ and $l = 1/e$, quenching diameter $d_q$ is given as $d_q = 2\sqrt{e\beta Nu}\delta_f$.

When $\beta$ around 10 and convective heat transfer of laminar flow in a cylindrical channel is considered, the quenching diameter is estimated approximately 20 times of the reaction-zone thickness. Numerical study addressing premixed flame propagation in two-dimensional channel by considering conduction to the channel wall and a prescribed Poiseuille flow [66] identified the quenching distance as the smallest channel width permitting flame propagation in cold tube. The ratio of the laminar flame thickness to the half channel width was equal to 0.13, which gives quenching Peclet number $d_q/\delta_f$ of 15.4. Quenching Peclet number was also addressed experimentally by using a large tube (i.d. 50 mm) at reduced pressure (2–60 kPa) [67]. Such flame is a model of ordinary ambient flame in narrow tube. Modified quenching Peclet numbers based on two kinds of characteristic flame length, such as measured thermal flame thickness and that estimated from the product of density and mass flow rate of stationary flames, were found to be distributed between 10 and 15.

To establish stable combustion in a small channel, stability managements are required. Promising method can be roughly divided into thermal and chemical approaches. The former include heat recirculation, heat shield, and high wall temperature. The latter is mainly attempted by catalytic reaction. Microscale combustion is often discussed in the context of combustion with heat recirculation, which is termed "excess-enthalpy combustion." It is collecting renewed and increasing interests due to its flame-strengthening effect as well as flame bifurcation, which results in weak flame, various novel instabilities and flame pattern formations. Weinberg and his co-workers [36,37,68] originally suggested the concept of excess-enthalpy combustion by using heat recirculating burners. It enables higher flame temperature than the adiabatic flame temperature, stable combustion for low calorific fuels and extended flammability limits [69,70] as well as sustaining combustion at small scales.

In the context of microscale combustion, the effects of heat recirculation on flame response were analyzed by Ronney and co-workers [71,72] and Ju and Xu [73]. U-shaped counter-current burner having inlet and outlet ports at one side and combustion chamber modeled as well-stirred reactor (WSR) at the other side was analyzed [71]. The model was formulated using three non-dimensional parameters, the Biot number ($B$), heat loss coefficient ($H$) and mass flux ($M$) in the

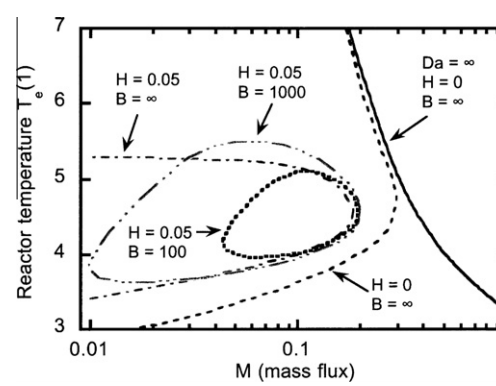


Fig. 10. Effect of mass flux on WSR temperature in the counter-current combustor for finite reaction rates at $Da = 10^7$ with $\Delta T = 1.5$ ($B$: Biot number and $H$: normalized heat loss coefficient).

channel. Figure 10 shows the effect of mass flux $M$ on the reaction temperature at the combustion chamber for adiabatic flame temperature rise at $\Delta T = 1.5$. In the adiabatic ($H = 0$) and infinite reaction rate ($Da = \infty$) case, flame temperature decreases as the mass flux $M$ increases. For finite reaction rate ($Da = 10^7$) case, there exists maximum mass flux $M$ which corresponds to the blow-off limit. For non-adiabatic ($H \neq 0$) case, lower mass flux limits in addition to blow-off limits exist when $B \neq \infty$, which indicates streamwise heat conduction in the dividing wall is dominant for the existence of the lower mass flux limit. This supports experimental facts, e.g., obtained by the coin-size Swiss-roll combustor [46,47].

The effects of external heat loss on flame speeds were conducted [74] using opposite propagating flame model. Opposite-propagation flame model enables explicit discussion on flame propagation under the heat-recirculation condition. Figure 11 shows flame speed as a function of external heat loss with heat recirculation. It is clearly indicated that the flame speed at extinction limit increased with increasing heat recirculation parameter $\chi$ In [74], significant extension of the flammable fuel-mass fraction by the heat recirculation was demonstrated as well. Opposite propagating flame model was used earlier for analyzing filtration gas combustion [75]. It is interesting to see common approach to examine microscale combustion and filtration gas combustion.

Advantages of the combustion with heat recirculation are extensively recognized and many studies addressed this issue. One-dimensional computation for hydrogen/air flame in a heat recirculating channel with detailed chemistry was conducted by Leach and Cadou [76]. Enhancement of stability limits and increase in burning rate were also shown. It concluded that combustor design that maximizes power density does not maximize efficiency. Two-dimensional modeling approaches for heat recirculating channel showed the existence of the flame bifurcation and significant effects of near-wall flame quenching and flame structure on flame speed [77], and the effect of wall thermal conductivity, external heat loss coefficient, flow rate and wall thickness of heat recirculating channel on lower and blow-off limits [78,79]. Optimal channel dimension was also discussed [80]. A heat recirculating combustor for a $CH_4$/air mixture having a straight main duct surrounded by annular porous inert media for preheating was presented [81]. Preheating annulus and insulation on the exterior surface reduced heat loss and offered potential to minimize surface heat loss in miniature combustion systems. Potential application for pre-vaporizing liquid fuels was suggested.

To examine the feasibility of stability management by high temperature wall, simple heat loss control by an external heater was employed for flame in a channel using methane/air mixture. Nearly flat flame in a channel with the inner diameter equal to its flame thickness was established under the reduced pressure [82]. The existence of flame with Peclet number unity, proves the theoretical implication by Daou and Matalon [66], showing no extinction limit for flame in an adiabatic channel, is still valid in such extreme condition without significant wall quenching effect.

### 4.2. Wall thermal effect

Combustion with heat recirculation is very effective for strengthen combustion. It also induces rich variety of flame dynamics due to flame–wall thermal interaction, as was also discussed in the context of mild and flameless combustion [83–85]. Weak flames and various instabilities result from flame bifurcation are discussed.

#### 4.2.1. Appearance of weak flames

Combustion in a narrow channel with heat recirculation was studied experimentally and theoretically [73,77]. One-dimensional quasi-steady flame propagation was modeled under the assumption of a constant wall temperature at upstream, uniform flow velocity, constant properties and a thin flame model. Figure 12 shows a model and results of analysis and experiments [73]. Normalized governing equations for gas and solid phases were solved and normalized flame speed $m$ ($=u - U_f$) as a function of normalized heat transfer rate $H$ was given. For the case at $u = 0$, normalized flame speed response exhibited stable-upper and unstable-lower solutions (not shown) which is similar to those in Fig. 11. On the contrary, normalized flame speed showed flame bifurcation at $u = 0.1$, exhibiting two stable flame branches and two extinction points for various wall thermal diffusivities (Fig. 12ii). This bifurcation leads to the appearance of weak flame, which is on the lower stable branch with very low propagation velocities. Pictures of $C_3H_8$/air flames in a mesoscale tube ($d = 5$ mm) are indicated in the same figure (iii) and showed flame propagation distance with time for normal (flame speed larger than 3.5 cm/s) and weak (flame speed less than 1 mm/s) flames. Flame bifurcation, resulted in weak flame in a heated microchannel, was independently reported [86,87]. Zamashchikov [88] experimentally showed the existence of very low-speed propagating flames in a tube which corresponds to weak flame. Two-temperature model successfully analyzed the experimental observation and showed the existence of weak flames [89]. Effects of Lewis number on flame in a channel including three-dimensional structures were also addressed [90].

Flame bifurcation was reported earlier for mesoscale particle-laden flame [91]. It is also noted

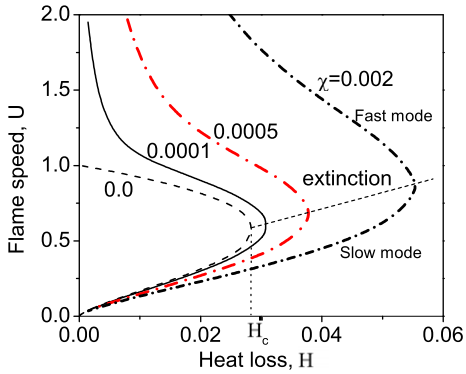


Fig. 11. Flame speed as function of external heat loss at different heat recirculation intensity.

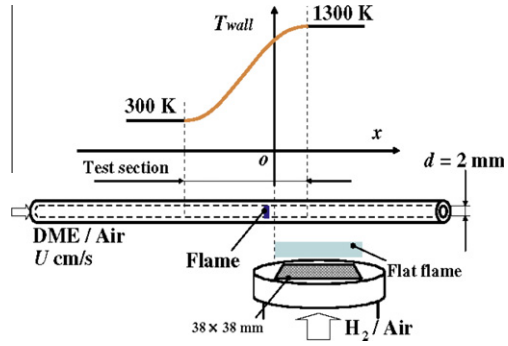


Fig. 13. A narrow channel with a prescribed wall temperature profile. Inner diameter of 2 mm, which is smaller than the ordinary quenching diameter, was chosen. Though hydrogen/air flat flame burner was shown as a heat source here, flat panel heaters were also used for earlier study.

that weak flame is in close connection with mild or flameless combustion, which is normally discussed in macroscale industrial applications.

*4.2.2. Repetitive extinction and ignition instability*

Microscale combustion was also examined using a narrow channel with a prescribed wall temperature profile along the flow direction [86,87]. A channel with the inner diameter smaller than the ordinary quenching diameter and an external heat source were used (electric heater or flat flame burner) for attaining a wall temperature profile (Fig. 13). Such approach was motivated during the development of disk-shaped Swiss-roll burners for general purpose heat sources [46,47], in which a narrow toroidal channel with temperature profile ramping from low to high temperature along the channel.

In addition to the normal flames, repetitive extinction and ignition instability was observed in a straight channel [86,87,92] and curved duct [93]. Figure 14 shows direct flame images of (A) normal flame and (B) flames with repetitive extinction and ignition. If flammable mixture experiences high enough temperature for auto-ignition, it is ignited and flame propagates to the upstream. Then flame leads to extinction due to the low-temperature in the upstream. After some time delay, re-ignition occurs at the high temperature region where previous auto-ignition started. This cycle is repeated regularly with frequencies from 10 to 100 Hz. It is termed "*flames with repetitive extinction and ignition* (FREI)" or "repetitive extinction and re-ignition instability" due to its nature. Clear image of the relevant oscillating flame in a curved duct, which exhibits repetitive extinction and ignition, is shown in Fig. 15 [93].

Figure 16 shows the details of dynamic flame motions. Figure 16a is a normal flame stabilized at a certain location in a channel. Figure 16b is a time history of reaction-zone location obtained by OH-filtered high-speed images, which is typical to FREI. In addition, one-dimensional pulsating flame (Fig. 16c), which exhibited regular periodical motions was observed. Its luminescence

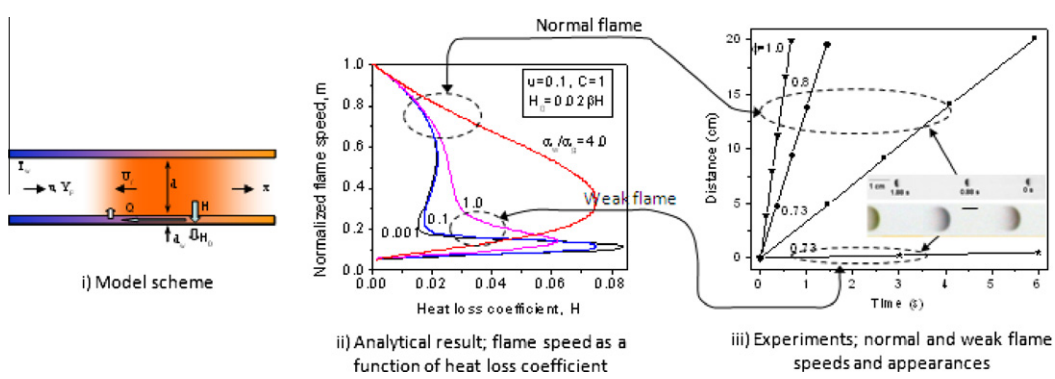


Fig. 12. Flame response with heat recirculation through thermally-coupled wall, $H$ and $H_0$: normalized convective heat transfer coefficients at the inner and outer surfaces of the channel wall, $C$: ratio of heat capacity of gas-phase to the solid structure, $\beta$: Zeldovich number, $\alpha_g$ and $\alpha_w$: thermal diffusivities of gas and solid structure.

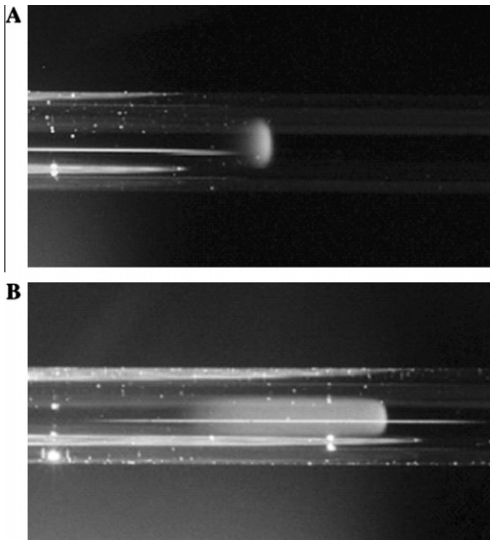


Fig. 14. Direct images of (A) normal flame and (B) *flames with repetitive extinction and ignition* (FREI) which were observed in a heated channel (a channel with a prescribed wall temperature profile). Mixture flows from right to left.

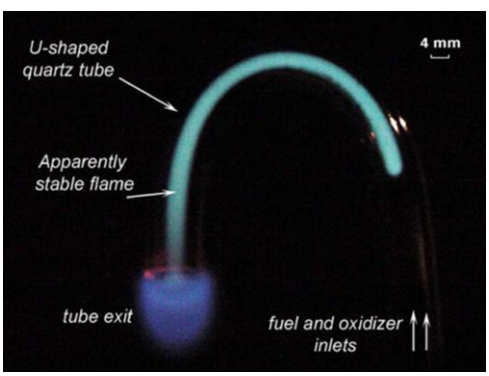


Fig. 15. Flame exhibiting repetitive extinction and ignition which was observed in a curved duct fuel rich condition.

remained almost constant and steady. Furthermore, flames which have the characteristics of both pulsating flame and FREI were also observed (Fig. 16d). The flames are characterized by two kinds of flame behaviors, which are a combination of the periodical motions of pulsating flame with small amplitude and repetitive ignition and extinction with large amplitude.

Theoretical study examined dynamics of flame in a channel with prescribed wall temperature profile [94]. Non-stationary flame behaviors as well as overall flame response are successfully captured analytically by proposing one-dimensional nonlinear evolutionary equation of flame front which considers flame front acceleration and rate of flame temperature variation [94]. Figure 17 shows stables flames at high and low velocity regions and FREI at middle velocity. Amplitudes of the FREI, that is, length between upstream and downstream turning points during FREI are clearly seen. Length of the FREI shortens when mixture flow velocity decreased toward the transition point from FREI (middle branch of steady-state solution) to weak flame regime (lower branch). This implies that weak flames are stabilized flame on an ignition branch. The theory provides very clear overall pictures of the experiments for a flame in a prescribed wall temperature profile [86,87].

In addition, two-dimensional computations for narrow tube captured oscillatory and chaotic flame dynamics [95], FREI-like limit-cycle phenomena close to its limit for self-stabilized flame with heat recirculation [78], and mixed and pure FREI [96]. Furthermore, two-dimensional computations with detailed chemistry and transport for fuel lean hydrogen/air flames ($\phi = 0.5$) in planar microchannels with prescribed wall temperature [97,98] identified much richer varieties of flame dynamics including open symmetric, asymmetric, oscillating, pulsating flames at higher flow rates as well as the details of FREI for the first time. Figure 18 shows the details of ignition/extinction behaviors shown by temporal variations of maximum temperature inside the channel, and two-dimensional distributions of OH (c–e) and $H_2$ (f–g) fractions at time instants $t_1$–$t_3$, for $U_{IN} = 1$ cm/s. At the moment of ignition of the incoming fresh mixture heated by the hot walls, confined flame propagates horizontally both upstream and downstream directions. The upstream propagating flame changes its configuration from curved to nearly flat flame and extinguishes in the upstream. The downstream propagating flame keeps its initial configuration in the remaining fresh mixture and extinguishes, finally. Such double ignition-kernels, propagating the opposite direction at the moment of re-ignition during FREI were independently observed, and thoroughly examined numerically and experimentally [99,100].

Detailed laser diagnosis on FREI were also conducted [101]. FREI were observed through $CH^*$ and $OH^*$ PLIF measurements and movements of these excited species during FREI are successfully captured for the first time. Besides, reaction-zone temperature measurement by OH two-line method showed flame temperature history from ignition to extinction during FREI, experimentally [102].

### 4.2.3. Lower limit of weak flame

For self-sustaining flame in a channel with heat recirculation via channel wall, lower limits of weak flame do exist due to the decrease of heat

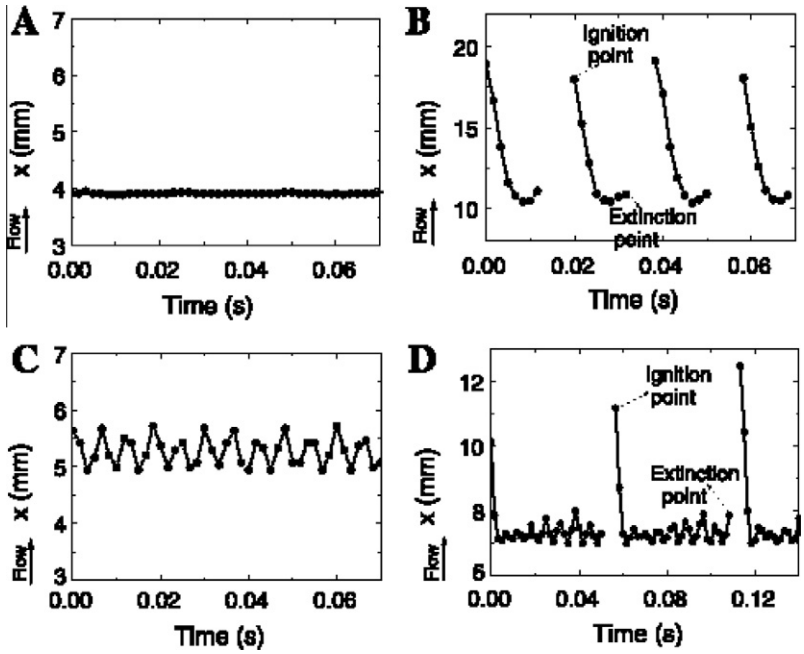


Fig. 16. Temporal histories of various flames of propane/air mixtures at an average mixture velocity of 30 cm/s and an equivalence ratio of 0.5; (A) stable flame in the case of maximum wall temperature, $T_c = 1320$ K; (B) flame with repetitive extinction and ignition, $T_c = 1130$ K; (C) pulsating flame, $T_c = 1270$ K; and (D) flame with a combination of pulsating flame and FREI, $T_c = 1200$ K.

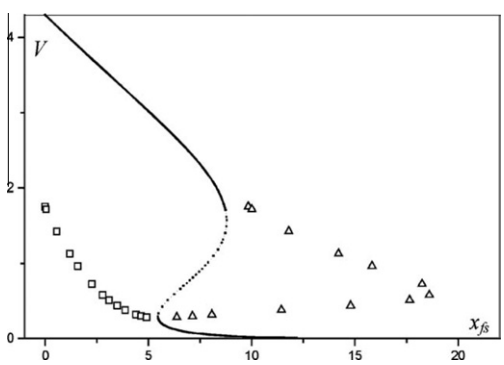


Fig. 17. Theoretical flame response for a narrow channel with a prescribed wall temperature. $V$ and $X_{fs}$ are flow velocity normalized by adiabatic flame speed and flame location. Overall flame response with two stable and one unstable branches were identified. Amplitudes of flame oscillation for FREI at middle branch are clearly predicted by nonlinear evolutionary equation of flame front. Triangles and squares are the upstream ('extinction') and downstream turning points ('ignition'), respectively.

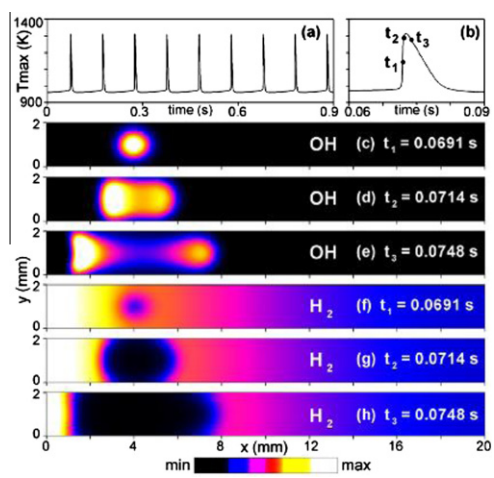


Fig. 18. Ignition/extinction mode at $U_{IN} = 1$ cm/s in the $h = 2$ mm channel: (a) temporal variation of the maximum temperature inside the channel; (b) expanded detail of (a); (c–e) and (f–h) two-dimensional maps of the $Y_{OH}$ and $Y_{H_2}$, respectively, at the three times $t_1$ to $t_3$ marked in (b). The minimum in the color-coded bar corresponds to zero for $Y_{OH}$ and $Y_{H_2}$, while the maximum to $Y_{OH} = 7.1 \times 10^{-4}$, $1.0 \times 10^{-3}$, and $6.0 \times 10^{-4}$, in (c–e), respectively, and to $5.14 \times 10^{-3}$ for $Y_{H_2}$.

generation with decreasing incoming enthalpy [73]. On the other hand, weak flame extinction of microscale flame in a tube with prescribed wall temperature does not likely occur, since tube wall temperature is somehow compensated by an external heater. This issue, the lower limit of weak flame, was recently addressed [103].

By the preliminary experiments, weak flame regime for $CH_4$/air mixture was identified at the mean flow velocity smaller than 3.0 cm/s. In the main experiments, although distinguishable weak flames were observed on or above 0.2 cm/s, no weak flame was observed at the mean flow velocity smaller than 0.2 cm/s, with even longer exposure time. This indicated the existence of the lower limit of weak flame in a prescribed wall temperature.

Figure 19 shows the difference between flame and wall temperatures in a weak flame regime. It shows temperature increase at the flame. The figure clearly indicates that the temperature differences decrease with the decrease of the mean flow velocity and the difference is almost zero at the mean flow velocity of 0.2 cm/s, where the wall temperature is around 1225 K regardless of mixture composition. Therefore, this temperature at the limit of weak flame can be considered as the ignition temperature of the given mixture based on the two characteristics of weak flame, (1) very small temperature increase and (2) stabilized flame on the ignition branch, which were inferred by the analysis (Fig. 17) [94].

Although ignition is usually considered to be transient phenomena, it was established as stationary propagating weak flame on the ignition branch with the help and temperature confinement by the prescribed wall temperature. Hence, ignition and limit of the weak flame propagation can be considered to be equivalent to each other at this extreme condition. This fact also indicates that the general ignition temperature of given mixture (at given residence time under the existence of diffusion), which is apparatus independent, can be identified with the position of the weak flame (local wall temperature) by this approach.

One-dimensional computations with detailed chemistry and transport properties was conducted [103] for more information. A steady-state flame code [87] with GRI-mech 3.0 was used. Energy equation with a convective heat transfer between gas and wall is expressed as:

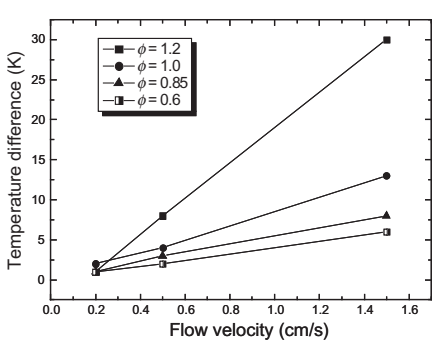


Fig. 19. Measured temperature difference between the flame and inner surface of tube wall at the flame position as a function of meat flow velocity.

$$\dot{M}\frac{dT}{dx} - \frac{1}{c_p}\frac{d}{dx}\left(\lambda A\frac{dT}{dx}\right) + \frac{A}{c_p}\sum_{k=1}^{K}\rho Y_k V_k c_{pk}\frac{dT}{dx} + \frac{A}{c_p}\sum_{k=1}^{K}\dot{\omega}_k h_k W_k - \frac{A}{c_p}\frac{4\lambda Nu}{d^2}(T_W - T) = 0.$$

Based on the fully developed flow in a circular tube for the case between constant wall heat flux and constant wall temperature, constant $Nu$ (Nusselt number) = 4 is assumed. Figure 20 shows the computed flame response to the mean flow velocity in a log-scale at equivalence ratio unity. The location of the CH peak is considered as flame location. Dashed line with open circles in the figure is the prescribed wall temperature. The computational flame response showed an ε-shaped curve, which has an additional lower velocity branch with the S-shaped curve in Fig. 17 [94]. Based on the stability analysis for the S-shaped curve, the lowest velocity branch is considered to be unstable. If this assumption is true, the existence of two stable and two unstable solutions in the four regimes can be inferred. Therefore, the existence of the lower limit of the experimental weak flame can be interpreted as the lower limit of the region (3) in Fig. 20.

Computational temperature difference between flame and wall at the flame position in weak flame regime [103] were quite similar to the experimental result shown in Fig. 19. Computed temperature difference becomes almost zero ($T_g - T_w < 1$ K) at mean flow velocity of 0.1 cm/s when wall temperature is 1230 K, which agrees very well with the experimental result (1225 K). However, this point locates in the middle of the lowest velocity branch (regime (4)) in Fig. 20. Since the boundary between the stable and unstable branches is unresolved, stability analysis should be conducted. It should be also noted that the extremely small temperature increase does not directly correspond to flame quenching in the low velocity regime; however, the conventional reaction with intense heat release no longer occurs in regime (4), even though the heat loss from the flame zone is compensated by the external heater.

If the thermal quenching does not occur at the lower limit of the stable weak flame, what is the possible mechanism of the extinction of weak flame? It is currently considered that the cause of such limit is related to the dominance of diffusive mass dissipation over the convective mass transfer in the extremely low velocity regime. Based on this assumption, the effect of diffusion on the lower limit of weak flame is examined computationally by comparing the convective and diffusive mass fluxes of OH in the case of high and low velocity regimes. OH was chosen because it is one of the key radicals for chain branching as well as its high diffusivity. Results indicated that diffusive mass dissipations from the reaction zone to both upstream and downstream directions were

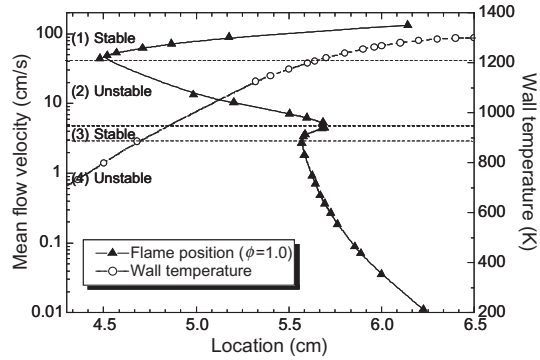


Fig. 20. Computed flame positions with mean flow velocity at the unity equivalence ratio.

significant close to the weak flame limit. Thus, lower limit of weak flame was attributed to this, which lowered the number of collisions and the rate of production of chain carriers, finally leads to extinction. Existence of the weak flame limit and its mechanism was also investigated under the reduced pressure [82,103].

*4.2.4. Pattern formation*

Spiral and other pattern formations, which are common in excitable media, were often observed in combustion systems [104–106]. Accordingly, pattern formations in microscale combustion have been also observed. Dynamic flames in straight, flat and radial channels exhibited various flame patterns with the parameter change. Periodic regular oscillatory phenomena made transitions and resulted in chaotic behavior or pattern formation.

Combustion in a straight tube (one-dimensional) with prescribed wall temperature profile was extended to a radial channel (two-dimensional) having a mixture supply tube at the center. A positive temperature gradient in radial direction from room temperature to around 850 K (prescribed radial wall temperature) was applied to a radial channel between two circular plates. Figure 21 shows rich varieties of instabilities and pattern formations, such as single, double, triple pelton-wheel-like flames, tri-brachial flame, spiral flame and traveling-wave flames as well as stable circular flame [107–109]. Each pattern can be transformed one to another with the variations of the parameters such as mixture flow rate, equivalence ratio and gap between two plates and those transitions were summarized as regime diagram [107]. As expected, most probable flame pattern in the regime diagram is stable circular flame when the gap between two plates is 2 mm. However, other patterns, such as single or double separated flames rotating around the center of the channel, which are termed as pelton-wheel-like flames [110], circular flames divided into some fractions where each fraction exhibiting tangential movement [111] are also observed. Spiral flames are observed at high velocity region at the gap between two plates larger than 2 mm [107,109]. The observed flame patterns are sensitive to the system parameters, particularly to the gap between two plates. For instance, no stable circular flames are observed when the gap is equal to 1.5 mm. Instead, traveling and pelton-wheel-like flames occupy 80% and 20% of regime diagram, respectively.

This implies that small scale combustion with radial flow geometry is likely to be suffered from analogous combustion instabilities. Accordingly, combustion completeness in the radial channel during the pattern formation is examined [108]. Combustion efficiencies estimated from the burned gas composition are 0.82–0.88 at double pelton flames whereas 0.48–0.82 at single pelton flame. Significant amount of unburned fuel and CO are flowing out without completing combustion in low combustion efficiency conditions. Difficulties in completing combustion at small scale combustor with two-dimensional geometries, such as MEMS micro gas-turbines, are inferred.

Mechanisms of pattern formations were investigated and few aspects were clarified to date. Inner and outer edges of the pelton and traveling flames are correlated with the locations of ignition and extinction of FREI phenomena in a straight tube based on experimental results [112]. Single/double pelton flames and spiral flames were numerically reproduced with two-dimensional transient computations [112] and their structure were examined. Clarifications of pattern formation mechanism and those flame structures are, however, difficult task and still ongoing.

Details of re-ignition in FREI on a straight channel is examined numerically and experimentally [99,100]. Splitting flames, which have dual reaction fronts, one propagating to upstream and another to downstream directions, are observed. This is due to the fact that re-ignition of FREI starts not at the edge of recharging fresh

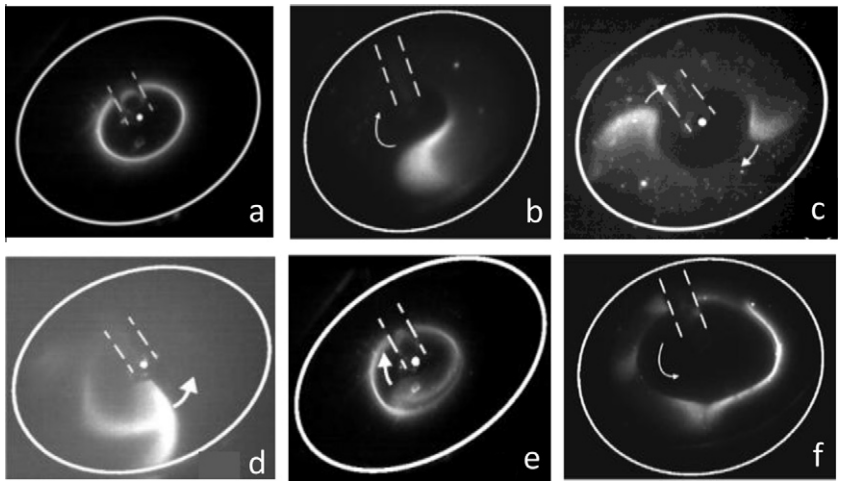


Fig. 21. Flame patterns observed in a radial microchannel with prescribed wall temperature profile under different conditions; (a) stable circular flame; (b, c) single and double pelton-wheel-like flames; (d) tri-brachial flame; (e) spiral flame; and (f) traveling flames.

mixture but at a certain point with small distance from the edge of the fresh mixture. Splitting flame behavior is playing key role for pattern formations in radial channel [112]. Splitting flame behavior is identified independently [98] by two-dimensional DNS for hydrogen/air mixture.

Rich varieties of flame patterns are also observed by two-dimensional direct numerical simulation with detailed chemistry and transport for hydrogen/air flames at atmospheric pressure in a rectangular channel between two parallel plates [97,98]. Temperature gradient along the flow direction which ramps smoothly from 300 to 960 K is given. Three kinds of channel heights, 2, 4 and 7 mm cases are examined for wide range of inlet mixture velocities from 0.3 up to 1100 cm/s. Because of this wide velocity range and wider channel height compared with the ordinal quenching distance of hydrogen/air mixture, rich varieties of flame patterns are obtained as well as comprehensive results which cover the conditions that are addressed by the existing experimental studies [87,93]. Four new kinds of flame regime, i.e., open steady symmetric flames, oscillating, asymmetric and chaotic flames are predicted at $U_{IN} > 75$ cm/s for the channel height of 4 mm, in addition to the three flame regime, such as weak flames, ignition/extinction and closed steady symmetric flames at $U_{IN} < 75$ cm/s.

Figure 22 shows the temporal variation of the integrated heat release rate and OH fraction contour of oscillating flame at $U_{IN} = 300$ cm/s. The oscillation of the integrated heat release rate is somehow periodic while oscillating flame at $U_{IN} = 100$ cm/s exhibits chaotic oscillatory behaviors. Two separated flames in the figure appear to be resulted from the interaction of the steady open symmetric flame with the chaotic cellular flame.

*4.2.5. Spinning flames*

Heat recirculation through tube wall or heat transfer in prescribed wall temperature geometry substantially modifies the balance between heat and mass transfer of flames in a tube. Thus, flame dynamics attributed to Lewis number effect is expected to occur. In fact, spinning flames are reported in a cylindrical divergent channel with heat recirculation [113] and a cylindrical straight channel with prescribed wall temperature [114].

Spinning flames in a divergent section of the convergent–divergent channel (Fig. 23) were reported [113]. Flame was stabilized in a divergent section as a weak flame under the strong thermal-coupling between flame and wall. Spinning flames was experimentally observed in addition to propagating, self-extinguishing and stabilized planar flames. After ignition of the mixture at the channel outlet, flame started to propagate toward section C through D, when the flame enters the section C, flame propagation speed decreases owing to the decrease of tube diameter where the local flow velocity increases. Finally, flame is stabilized in a certain location in the section C as a planar stable ($Q = 3$ cm$^3$/s) or spinning flames ($Q = 5$ cm$^3$/s) at $\phi = 1.5$. Some time delay existed for the start of spinning motion due to the time requirement for the flame–wall thermal-coupling was established. Spinning direction was found to be random. There exists a critical flow rate, above which spinning flame exists and below which flame is stable both methane/air and propane/air flames. For methane flames, there is a quenching limit at low flow rates. Close to the

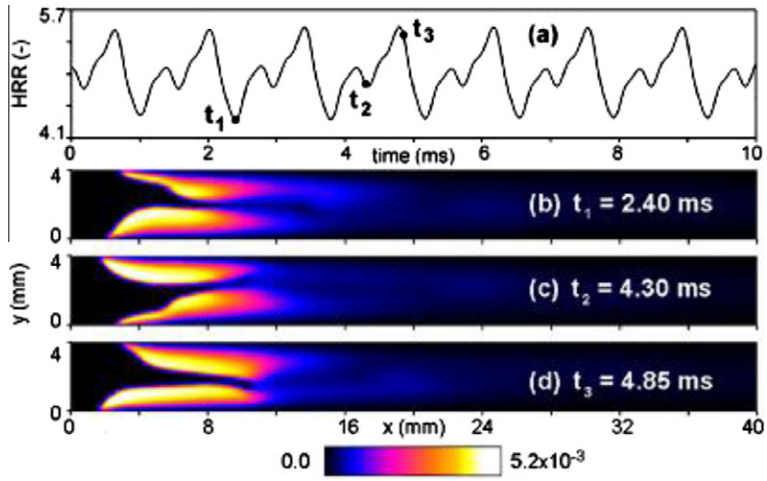


Fig. 22. Oscillating flames at inlet mixture velocity at 300 cm/s in the 4 mm height channel; (a) time history of integrated heat release rate and (b)–(d) OH fractions at the time instants $t_1$ to $t_3$.

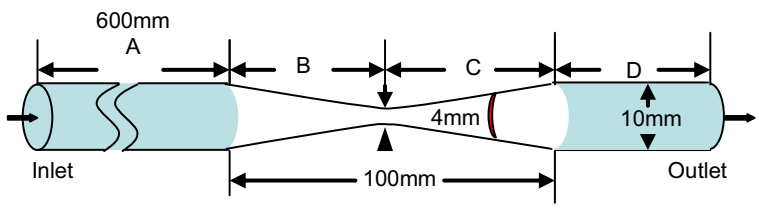


Fig. 23. Scheme of convergent–divergent channel.

stoichiometric conditions, propane flames with low flow rate pass thorough the throat without extinction. It was found that the spin frequency is almost proportional to the average flame speed but not depend on the flow rate. Spinning flames were observed for both methane and propane flames at lean and rich conditions regardless of the mixture Lewis numbers. In a conventional theory on diffusive-thermal instability [115], flame instability appears at mixture Lewis numbers larger than the critical value. Therefore, the current results seem to be inconsistent with the theory of diffusive-thermal instability of non-adiabatic flames. However, it was pointed out that the balance between heat and mass transfers are different from that obtained by mixture Lewis number due to the heat recirculation through tube wall. Hence, effective Lewis number, which accounts for the heat transfer through the tube wall, was estimated to examine the condition for the onset of flame spinning. Figure 24 shows effective Lewis number as a function of normalized tube diameter which was theoretically estimated by matching the flame speed with and without heat recirculation [113]. Significant increase in the estimated effective Lewis number at small normalized diameter readily results in exceeding the critical non-dimensional Lewis number $\beta(Le-1) = 21/2$ for adiabatic and 18/5 for limiting cases in the conventional theory, if the Zeldovich number assumed to be around 10. Thus, it can be concluded that the current spinning flames are induced by the increase of the

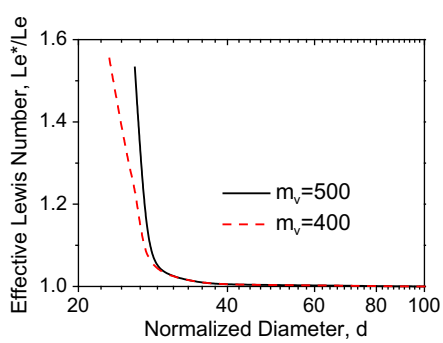


Fig. 24. Effective Lewis number estimated from theoretical analysis by matching the flame speeds with and without heat recirculation for methane–air mixture. $m_v$ is volumetric flow rate normalized by the product of adiabatic flame speed and the square of flame thickness.

effective Lewis number due to the heat recirculation. Two curves with different flow rates in the figure clearly explain the existence of the critical flow rate above which the flame spinning is being observed.

Very recently, three-dimensional numerical simulation with detailed chemistry and transport on premixed hydrogen/air flames ($\phi = 0.5$) in cylindrical tubes (inner diameters of 1.0 and 1.5 mm) with prescribed wall temperature profile was conducted [114]. In the wider tube, spinning flames (Fig. 25) as well as steady mild combustion, oscillatory ignition/extinction, steady closed and open axisymmetric flames, and steady non-axisymmetric flames were identified. Coexistence of the spinning and axisymmetric flames was further discussed based on axisymmetric computations to elucidate the nature of the observed transitions in the wider tube. Fourier analysis during the transitions showed that a unity wave-number azimuthal mode plays a dominant role in the transitions. Significant spinning-frequency dependence on the flow velocity, which contradicts those found in a divergent tube [113], was reported. The success of three-dimensional DNS with detailed chemistry shows that the microscale combustion is one of the perfect platforms for the interactive developments of multi-dimensional DNS, comprehensive theory and experiments using modern diagnosis.

### 4.3. Wall chemical effect

#### 4.3.1. Catalytic combustion

In microscale applications, catalytic exothermic reaction is advantageous because of the large surface-to-volume ratio, immobile heat release zone, no quenching distance, moderate reaction temperature and clean emission. Some new fundamental aspects of catalytic combustion, design, operation, modeling and diagnosis which are

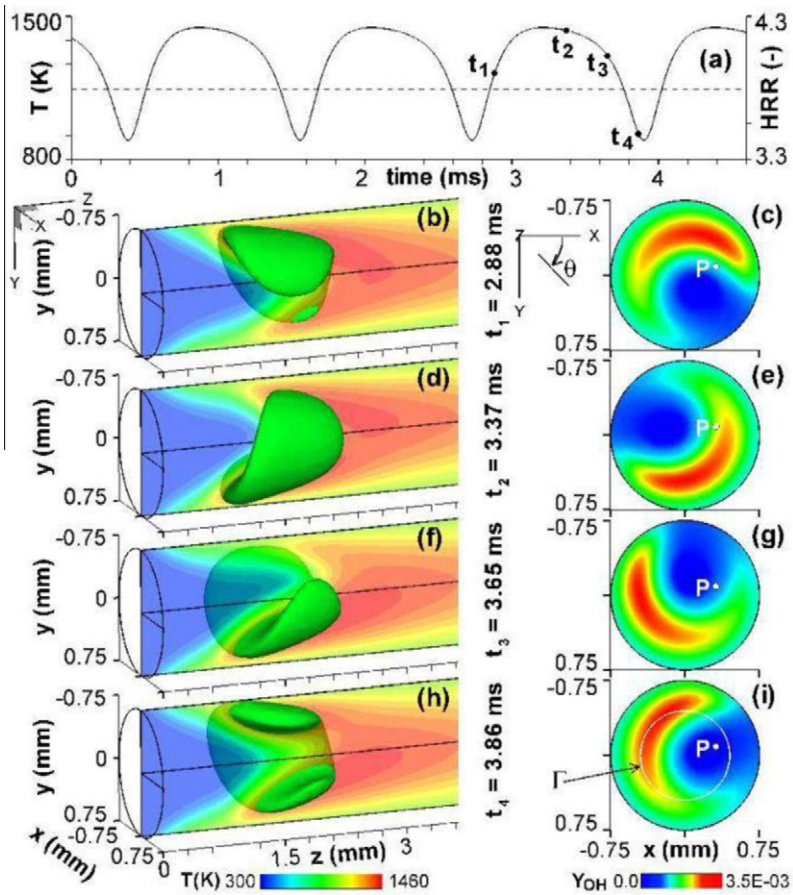


Fig. 25. Clockwise spinning flame at the flow velocity of 150 cm/s in the 1.5 mm tube: (a) time history of the temperature (solid line) at the reference point P at $(x, y, z) = (0.3, 0.1, 2.06)$ mm, and integral heat release HRR (dashed line); (b, d, f, h) iso-surfaces of $Y_{OH} = 1.7 \times 10^{-3}$, and temperature iso-contours on the $y$–$z$ plane at the four times $t_1$ to $t_4$ shown in (a); (c, e, g, i) iso-contours of $Y_{OH}$ on the $z = 1.5$ mm plane at times $t_1$ to $t_4$.

newly highlighted for microcombustion application are briefly overviewed.

*4.3.1.1. Stability and flammability limits of catalytic combustion.* Flame stability and flammability limits are one of the key fundamental subjects of combustion science. Thus, "flammability limit" of catalytic combustion was addressed using Swiss-roll burners [116] for very wide range of physical parameters. Their principle findings are capability of low-temperature catalytic reaction and flammability limit boundaries which are unsymmetrical to the stoichiometry and extremely wide toward fuel rich condition. A 3.5-turn square spiral counterflow Swiss-roll burner with $70 \times 70 \times 50$ mm tall having 3.5 mm channel width was constructed and bare platinum strips with approximately 30 $cm^2$ in total area were placed at the center of the burner. Propane/air mixture and other diluents were used and both catalytic and gas-phase combustion were tested in the experiments. Figure 26 shows the lean and rich extinction limit boundaries of catalytic and gas-phase combustion on Reynolds number – equivalence ratio plane. Notable feature in the figure is very wide self-sustained boundary of catalytic combustion, which has reached down to very small *Re* close to unity, being unsymmetrical to the stoichiometric condition. Its difference with that of gas-phase combustion is remarkable especially in low *Re* and fuel rich conditions. For example, gas-phase reaction cannot be sustained at $Re < 40$, while catalytic combustion with $NH_3$ conditioned catalysis can be sustained at $Re \simeq 1$, where maximum temperature record at the burner center is very low (350 K at $Re = 1.2$.)

It is also noted that at $Re < 15$, lean extinction limit of catalytic combustion is at richer side of the stoichiometric condition. Rich limit of catalytic combustion was very rich, e.g., $\phi_{lim} > 40$ at $Re = 15$. This fact suggests that catalytic flammability limit is totally governed by the different mechanism with that of gas-phase combustion.

Maximum burner temperatures as a function of *Re*, along the extinction limits in Fig. 26 was recorded and the lowest temperature for self-sustaining catalytic combustion is 350 K at $Re = 1.2$, when $NH_3$ conditioned catalysis was used. Meanwhile, the lowest temperatures for self-sustaining gas-phase reactions are far above 900 K. It is emphasized again that those remarkably low self-sustaining temperatures for catalytic combustion at "lean" limits were accomplished at the rich side of the stoichiometry (Fig. 26).

Capability of low-temperature self-sustaining catalytic combustion at the rich side of stoichiometry has been reported earlier by using simple catalytic channel [117]. Figure 27 shows the lowest self-sustaining temperature curve, which is discontinuous at stoichiometric condition and required temperature drops by almost half there, demonstrated by varying fuel-to-oxygen ratio in a methane/nitrogen/oxygen mixture. The figure shows extinction limits and required temperatures for self-sustaining reaction of $N_2$-diluted methane/oxygen mixtures in a non-adiabatic, 1 mm diameter catalytic channel at representative operating condition of mixture velocity = 0.06 m/s and heat transfer coefficient from the outside of the channel wall to the ambient is 5 $W/m^2$ K. The limit temperatures are higher than 1200 K in fuel lean condition while those at fuel rich condition are lower than 800 K and decreasing down to 600 K as shown in the figure. This high/low limit temperature at fuel lean/rich side and its discontinuity at the stoichiometry is interpreted in terms of inhibition of methane absorption by O(s) surface coverage [118], which drastically differs at lean and rich sides of the stoichiometry.

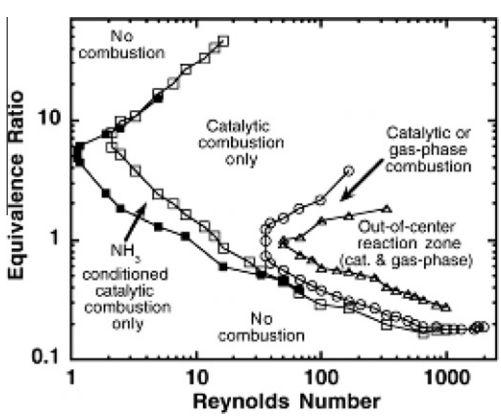


Fig. 26. Stability boundary of catalytic and gas-phase combustion on Reynolds number – equivalence ratio plane. Reynolds number is defined by the area-averaged inlet mixture velocity at the standard state. Swiss-roll burner is made of Inconel.

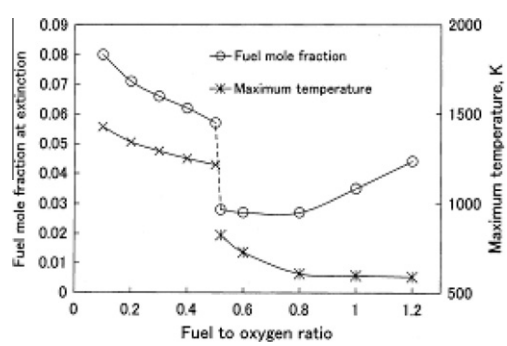


Fig. 27. Lower self-sustaining limits of fuel concentration and operation temperature for $N_2$-diluted methane/oxygen mixtures in a Pt catalyst channel (1 mm in diameter) at the averaged inlet mixture velocity 0.06 m/s under heat loss condition. Heat transfer coefficient between outer wall to the ambient is 5 $W/m^2$ K.

Additional advantage of fuel rich operation for limiting formation of metal oxides is discussed in [119]. Metal oxides are often volatile than the reduced metal and loss of Pt in the form of $PtO_2$ is a significant mechanisms for catalyst deactivation under fuel lean operation. Under fuel rich conditions, however, volatility of the catalyst is not a significant deactivation mechanism, allowing long catalyst life and broad choice of catalyst materials, without constraints regarding oxide volatility.

*4.3.1.2. Interaction/transition between gas-phase and surface reactions.* Interaction/transition between surface and gas-phase reactions (hetero/homogeneous reaction interaction) is important for understanding the operation boundary of catalytic combustion. It was addressed by several studies [120–122] for macroscale applications of catalytic partial oxidation and $NO_x$ reduction and use of high temperature heat exchangers [123].

Stability diagram with evident extension of stability limits due to gas-phase reaction was presented for $10 \times 1$ mm Pt-coated catalytic plane channel, computationally [124]. Heat conduction in the wall (wall thermal conductivity, $k_s$), heat loss from outer wall to the ambient (heat transfer coefficient, $h$), and inlet flow velocity, $U_{IN}$ were parameterized using methane/air mixture of $\phi = 0.4$ at inlet temperature 600 and 700 K. Figure 28 [124] shows stability diagram constructed by varying external heat transfer coefficient and the inlet flow velocity when wall thermal conductivity, $k_s = 2$ W/m K. Dual-limit behavior was observed. Blow-off limits obtained with gas-phase chemistry were extended both for $T_{IN} = 600$ K and 700 K at 5 bar. These limit extensions are interpreted by gas-phase homogeneous reaction occurred along the center of the channel, at slightly downstream of the region of surface reaction as shown by streamwise profiles of methane conversions in gas-phase and at the catalyst surface.

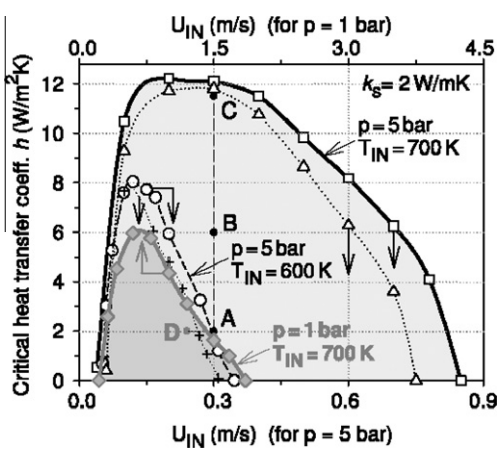


Fig. 28. Stability diagrams of critical heat transfer coefficient and inlet velocity at $p = 5$ bar (squares: $T_{IN} = 700$ K, triangles: $T_{IN} = 700$ K without gas-phase chemistry, circles: $T_{IN} = 600$ K, crosses: $T_{IN} = 600$ K without gas-phase chemistry), and at $p = 1$ bar (filled diamonds: $T_{IN} = 700$ K). The stable regimes for the 700 K cases are shown by the shaded areas.

Similar tendencies containing interaction between gas-phase and surface reactions were reported for hydrogen/air mixture at ambient pressure and temperature [125]. Both gas-phase and surface reactions are computed in a 10 mm × 1 mm-diameter thermally thin cylindrical Pt channel. Convective and radiative external heat losses were considered. Figure 29 [125] shows computed temperature and OH concentration distributions for stoichiometric hydrogen/air mixture at an inlet flow velocity of 2 m/s by three reaction modes: both gas-phase and surface reactions, gas-phase reaction-alone and surface reaction-alone. Though comparatively rapid temperature rises close to the channel inlet were attained in all cases, most prompt temperature increase was seen in gas-phase reaction-alone case, not in both gas-phase and surface reaction case, which implies that gas-phase reaction is inhibited by catalyst.

This tendency can be more evidently seen from the absence of OH in the vicinity of channel inlet, especially in a region close to catalyst wall for both gas-phase and surface reaction case. OH concentration is higher at wall vicinity for surface reaction-alone case than that for both gas-phase and surface reaction case, and more significantly, OH are only seen in the vicinity of catalyst wall for surface reaction-alone case because chemical reaction only occurs on the wall in this case. In a single-channel catalytic micro burner, self-ignition, gas-phase combustion in parallel with catalytic combustion and audible "whistle" together with temperature bifurcation were also identified [126].

The effect of channel diameter on the interaction between gas-phase and surface reactions is also essential. Since characteristic diffusion time scale is proportional to the square of channel diameter, the decrease in channel diameter leads to weaker gas-phase reaction through the fuel depletion by surface reactions. Hence, OH concentration along the center axis in gas-phase shifts upstream and sharply decreases with the decrease of tube diameter. On the contrary, blow out of gas-phase reaction is easy to occur when channel diameter is too large, where promotion of gas-phase reaction through high temperature and radicals produced by surface reaction is scarcely expected. Eventually, optimum channel diameter exists for intense gas-phase reaction as shown by blow-off limits as a function of channel diameter [125]. By summarizing this, overall picture of gas-phase and surface reaction interaction divided

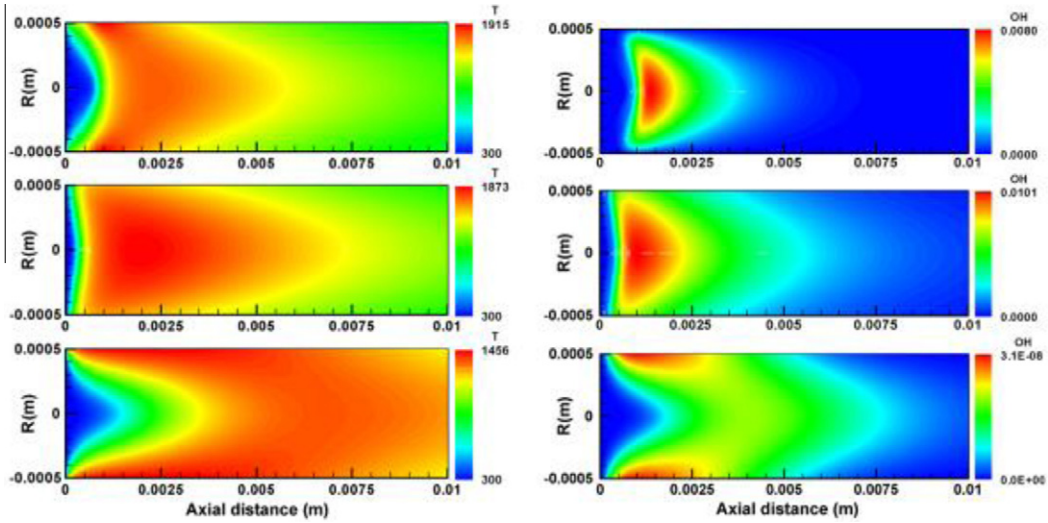


Fig. 29. Computed temperature (left) and OH concentration (right) distributions for three different modes: upper: both homogeneous and heterogeneous reactions, middle: homogeneous reaction-alone, and lower: heterogeneous reaction-alone.

into three regions was shown in Fig. 30. Gas-phase reaction exceeds surface reaction in the region I, gas-phase reaction is inhibited by surface reaction to some extent but they both co-exist in region II, and surface reaction is dominant in region III [125]. Line A in the figure corresponds to the blow out limits of gas-phase reaction. Above the line B, gas-phase reaction is inhibited and has no effects on the phenomena. Since hydrogen/air mixture is lead to light-off even at room temperature, no lower velocity limit exists

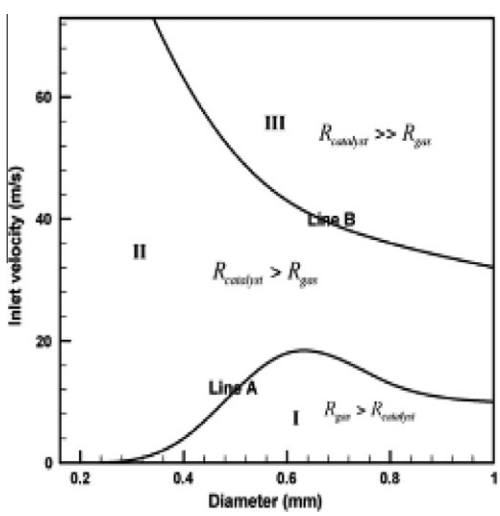


Fig. 30. Regime of gas-phase and surface reaction interaction. region I: gas-phase reaction dominant; region II: gas-phase reaction inhibited by surface reaction but they both co-exist; and region III: surface reaction dominant.

in the figure. The existence of upper limit of surface reaction for hydrogen/air mixture was not discussed. For mixtures with moderate surface reaction rate, e.g., methane/air mixture, have both lower and blow-off limits [117,124].

For comprehensive understandings on the limit behavior of catalyst together with interaction of the hetero/homogeneous reactions, further investigations at lower catalyst temperatures are required. Specific characteristics of catalytic combustion induced by discontinuous lowest self-sustaining temperature at the stoichiometry, competing with hetero/homogeneous reaction interaction would lead to wide variety of new phenomena in this area.

*4.3.1.3. Ignition and low-temperature catalytic reaction.* Ignition or light-off of catalytic reaction is another key subject of catalytic microcombustion system. Significant role of adsorption–desorption on catalytic ignition was presented in one of the early numerical studies [118], though it was not addressed for micro power generation. In brief, fuel rich conditions for hydrocarbon fuels and fuel lean conditions for hydrogen are advantageous for low temperature ignition of catalytic reactions since catalytic ignition is governed by the availability of sufficient uncovered surface sites. Detailed earlier studies on catalytic ignition were also conducted [127].

Such characteristics motivated various ignition strategies for catalytic microcombustion systems as well as self-ignition of hydrogen/oxygen mixture at room temperature.

Hydrogen assisted self-ignition of methane/air [128] and propane/air [129] mixtures were investi-

gated. Norton et al. [129] pointed out that concept of hydrogen assisted self-ignition of hydrocarbon can be applied to catalytic microscale devices without ignition sources. More practical approach was conducted to identify the effective ignition strategies for simple catalytic channel with heat conducting wall numerically [130]. Multi-segment catalyst disposition is also examined numerically for hydrogen/air mixture in a 1-mm height micro channel [131].

Catalyst pretreatment is common strategies for achieving desirable catalyst ignition characteristics. Catalyst activation at lower temperatures for hydrocarbon fuels were attempted by using $NH_3$ pretreatment [116]. This technique was applied for low-temperature catalytic "plastic" combustor system which produces electric power [132] with so-called micro heat engine [133]. The micro heat engine converts thermal power to mechanical power through the expansion and compression of a two phase working fluid. Then the mechanical power is converted into electrical power through the use of a thin-film piezoelectric generator [133]. Room temperature ignition for DME at platinum pellet catalytic reactor is also achieved by pure DME feed followed by mixture supply [134].

### 4.3.2. Radical quenching

The effects of radical quenching on flame speed and extinction are considered to be significant in microscale combustion due to the large surface-to-volume ratio. The wall chemical effect has been discussed based on interfacial concentration and temperature discontinuities at the wall [135] in microscale combustion system. Radical quenching at wall [78,136,137] and non-equilibrium slip in catalytic combustion [135,138] were addressed to date. Physically, the former is considered that radicals adsorb on the wall where they recombine by the surface reactions through surface diffusion to form stable molecules that could desorb to the gas-phase. The radical removal rate is modeled using the sticking coefficient, describing the probability of an impinging molecule to stick to the surface and the probability of a radical encountering an empty active wall site [139]. The limiting case of zero sticking coefficient corresponds to an inert wall, while a sticking coefficient of one corresponds to the fastest possible radical removal. A study on stagnation flow [139] found that radical recombination was exothermic and its released heat can have a net stabilizing effect on the flame under certain conditions, while radical removal retards combustion under isothermal conditions and can cause extinction.

Two-dimensional computations for preheated methane/air combustion in a non-adiabatic straight channel with detailed chemistry and transport focusing on the gas-surface interfacial phenomena were conducted [78,136]. Figure 31a shows the mole fraction of methane for a quenchless wall (solid line) and for a fast radical quenching wall (dashed line) for a 100 μm radius channel. Figure 31b shows the corresponding wall temperature. The figure shows radical quenching can result in flame extinction. It is also seen that the rate of heat released by radical recombination at the wall is insufficient to compensate for the rate of heat loss through the wall. Therefore, the kinetic mechanism clearly dominates over the thermal process.

For qualitative but general understanding, numerical study using three-step skeletal kinetic mechanism, consists of chain-initiation, branching and termination process, as well as radical quenching on the wall surface was conducted [137]. Steady flame propagation in a two-dimensional mesoscale channel was considered. The effects of channel widths, diffusivity of radicals and radical quenching coefficient on the flame speed were discussed by using two different values of crossover chain-branching temperature. Contours of fuel and radical mass fractions (not shown here) of stabilized flame indicated the existence of significant amount of radicals near the wall for the case without radical quenching, while the radicals near the wall is consumed due to the radical quenching at the wall for the case with radical quenching. Figure 32 shows the effects of radical quenching on the flame speed at different wall temperatures. Radical quenching does not play significant role at large channel widths. When the channel width decreases, radical quenching starts to reduce the flame speed. At higher wall temperatures, radical quenching on the wall

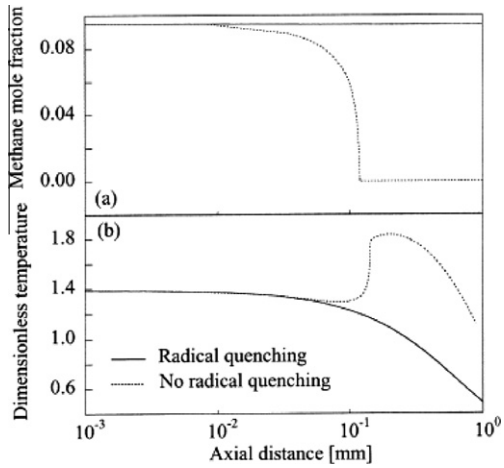


Fig. 31. Mole fraction of (a) methane and (b) dimensionless temperature (normalized by 1273.15 K) along the length of the reactor for quenchless walls (solid lines) and quenching material (dashed lines). Temperature discontinuity is not taken into account in these simulations. External heat transfer coefficient at 4.187 $W/m^2$ K.

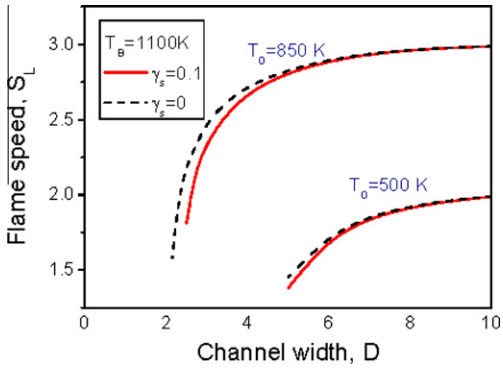


Fig. 32. Effects of radical quenching on the channel width and inlet temperature $T_0$ on flame speed (normalized). $T_B$ and $\mu_s$ are employed chain-branching temperature level and radical quenching coefficient. Radical quenching is significant at higher temperature.

becomes more significant since flame can be sustained at smaller channel and the radical concentration is higher. In addition, radical quenching lead to narrower flammability limit via decreasing the flame speed.

Feasibility study of hydrocarbon combustion in a small burner was experimentally conducted by paying special attention to thermal and radical quenching [140]. Several wall materials which are not expected to trap radicals are selected based on an earlier analysis [141]. By using the gap formed by those materials, quenching test was conducted with a standard methane/oxygen torch. The wall temperature was controlled by heating or cooling the material blocks to maintain an inside wall temperature at constant. Results show that quenching distance did not depend on wall materials when the wall temperature is below 500 K whereas those at wall temperature near 1273 K are strongly dependent on the wall materials. It was concluded that the thermal quenching was dominant at low-temperature wall while the radical quenching was dominant at high temperature wall. Effects of thermal and chemical surface–flame interaction on flame quenching were experimentally investigated at KAIST [142]. Reactive and inert test plates with three kinds of materials, such as stainless steel, alumina and quartz are chosen for the panels organizing narrow gap. Nearly identical quenching distances are demonstrated at wall temperature at 500 K regardless of wall material and pretreatment, while material and pretreatment sensitivities are confirmed at wall temperature at 1100 K.

### 4.4. Diffusion-controlled phenomena

#### 4.4.1. Diffusion-flame instability

Novel cellular-type diffusion-flame instability in flow direction was observed [143] in a 30 × 5 × 0.75-mm Y-shaped rectangular channel in a polycrystalline alumina burner as shown in Fig. 33. Fuel and oxidizer were supplied from the Y-shaped separated inlet ports at the bottom of the burner and exhaust gas was ejected from the outlet at the top. Observed flame structures were found to be sensitive to fuels employed ($H_2$, $CH_4$ and $C_3H_8$). The flame instabilities were only seen for $CH_4$ and $C_3H_8$ cases, while they were not observed for $H_2$–$O_2$ case except the case with He dilution. The number of flame cells in the burner was dependent on flow rates and overall stoichiometry.

To elucidate the mechanism of diffusion-flame instability in a mixing layer in microscale channel, Xu and Ju [144] examined the dynamics of diffusion flame in a mesoscale channel with elevated temperature wall, experimentally and analytically. By the study, structure of flame streets in the mixing layer was shown to be multiple triple-flames. The number of flamelets and the flame distance in the mixing layer depends on the flow rate, wall temperature, and fuel diffusivity. Figure 34 shows multiple flamelets at different wall temperatures. At higher wall temperatures, the heat loss from flame to the wall is reduced and the flame temperature will increase. Thus, the flamelets can stabilize at more diluted conditions and the flame separation distance becomes shorter. This effect surpasses the flow velocity increase by the thermal expansion for methane case. The size of the flamelets was proportional to the width of the mixing layer. An analytical scale model qualitatively explained the mechanism of flame streets and reproduced the flame separation distance. At lower flow rate and lower temperature region, an unsteady bimodal flame regime was also identified in addition to the flame streets.

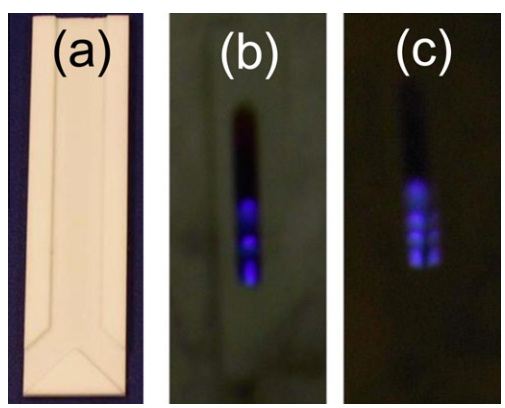


Fig. 33. Diffusion-flame instabilities observed in (a) 30 × 5 × 0.75-mm Y-shaped rectangular channel in an alumina burner. Flame images of (b) three flame cells formed with 100 sccm $CH_4$/200 sccm $O_2$, and (c) four flame cells for 100 sccm $CH_4$/130 sccm $O_2$.

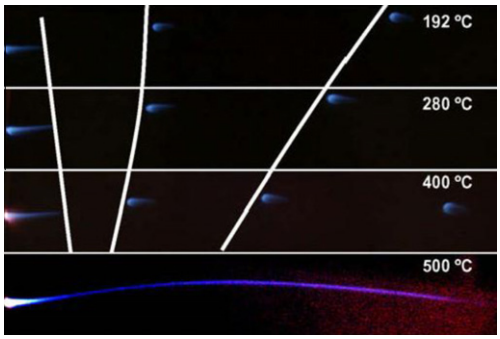


Fig. 34. Photographs of flame streets in a fuel/oxidizer mixing layer in a microscale channel at different wall temperatures. At wall temperature 500 °C, diffusion flame branch does not quench and all flamelets merge, leading to a single triple flame structure.

*4.4.2. Microscale jet flames*

For examining diffusion limited phenomena, microscale hydrogen-jet diffusion flames formed by microscale tubes ($d = 0.2$ and 0.48 mm) were investigated with UV Raman scattering coupled with LIPF technique [145]. Point measurements of temperature, major species concentrations ($O_2$, $N_2$, $H_2O$ and $H_2$), and absolute hydroxyl radical concentration (OH) were conducted. Two-dimensional OH image shows that a spherical flame with 1-mm radius was established at 0.2-mm tube. Computation showed that the flames are in the convection–diffusion controlled regime because of low Peclet numbers.

A theoretical study on the extinction limit of microscale jet diffusion flames conducted by Kuwana et al. [146] showed that the mixture-fraction gradient at the flame tip greater than a critical value leads to flame extinction.

## 5. Summary and future of micro and mesoscale combustion

Micro and mesoscale combustion engines, turbines, thrusters, reactors and power generators with various operation principles were introduced. Though the developments are still not at fully matured stage, sufficient feasibilities were demonstrated through the technical challenges. As noted at the beginning, combustion-based small scale energy devices possess broad potential applications not only as electrical power source but as heat and mechanical power sources. Thus, scale-down approaches of the existing macroscale IC engines, turbines as well as devices with specific operation principles which are advantageous in small scales have bright future-outlook. Various microscale combustors with/without energy conversion devices, micro-thrusters are also promising as mechanical power and heat sources. Humanoid robots, unmanned vehicles, powered wheelchair, etc., which definitely need portable power sources, will have growing markets in the future. By considering the average power consumption of human beings are slightly larger than 100 W, $CO_2$ emission from the combustion-based smaller power generators with the same power levels may not have serious environmental impacts. It is noted that application to micro power generation is not for any global energy or $CO_2$ issues but for the future value-added, reasonable products which utilizes high energy density of hydrocarbon fuels.

Flame bifurcations induced by wall thermal effects exhibited weak flames and dynamic oscillating and spinning flames. Experimental, computational and theoretical studies stimulated each other and disclosed rich variety of novel micro/mesoscale flame phenomena. Low-temperature catalyst operation, interaction between gas-phase and surface reactions and radical quenching gained increasing interests through wall chemical effects. Novel diffusion-controlled phenomena were observed and analyzed.

In addition to them, what we have learned from fundamental microscale combustion are expected to be applied for new concept combustion technologies. By the knowledge of flame bifurcation, weak flame and extinction and ignition instabilities, we may establish very slow and/or spatially distributed heat sources for further energy conversions. Catalytic combustion with extremely low-temperature will be technically useful, and also may lead to totally unexpected catalyst usage. Knowledge on predominant diffusion also might lead new applications. For summary, some new trials and future directions are discussed.

*5.1. Microscale engines and power generators*

Future hybrid and electric vehicles are requiring power sources for driving compressor, generator, pump and even heat sources for air conditioning and adsorption heat pump system. Therefore, further developments of scale-down versions of existing engines, such as SI, and compression–ignition engines, gas-turbines, rotary and free-piston engines, Starling engines as well as efficient, reliable, microscale combustors with clean emissions and robust controls are required. Low temperature operation using weak flame or dynamic flames may be possible for low power density sources due to higher energy efficiency using slower process.

Combinations of combustors/reactors with energy conversion devices are also promising. The latest development on 1-cm-size power generators using a tube-type catalytic combustor and conventional thermoelectric devices marked 3.52% of total efficiency and energy density

420 W h/kg [147], which is nearly 2.5 times of Li-ion batteries. Micro power generator with TPV device or other conversion principles are also expected to be invented for new applications.

### 5.2. Micro reactors for fuel reforming

Microreactors for fuel reforming using thermal energy based on small combustion devices are promising. Heat and mass transfers, and reaction processes can be optimized through the precise process control rather than the macroscale counterparts. Recently, the conversions of wet ethanol, or ethanol that has not been fully distilled or dehydrated, to syngas were investigated using combustion in porous media [148], and shown that wet ethanol to be a promising biological source for hydrogen. Design challenges for such direction using flexibility of microcombustion are plausible.

### 5.3. Microscale combustion for fuel characterizations

Knowledge on weak flame and increasing demand for anti-knocking fuels for future HCCI engines motivated fundamental study of ignition and combustion characteristics of practical fuels using microcombustion. By using microscale combustion in a channel with a prescribed wall temperature profile, DME [149], *n*-heptane [150] and gasoline-PRF are tested in the channel and multiple weak flames, i.e., stationary multi-stage oxidations, which correspond to the ordinary transient multi-stage ignition phenomena, were identified. Information on low temperature reaction, contributions of multi-staged heat releases and their pressure dependencies will be easily investigated. It may contribute small, low-cost apparatus for fuel characterization and indexing.

### 5.4. Microscale combustion as a standard platform for fundamental studies

Due to its small scale nature, microscale combustion can be used as a standard benchmark platform for the development, verification and improvement of comprehensive theories, detailed computations with full chemistry and cutting-edge diagnosis and measurements. In fact, DNS on hydrogen-fueled microscale combustion has identified spinning flames as well as reproducing experimental findings [114]. The latest DNS study on micro swirling-flow combustor clearly captured instantaneous contour surfaces of the second invariant of velocity gradient tensors, together with heat release rate contours, which clearly showed flame-vortices interactions [151]. DNS on higher hydrocarbons with huge chemistry are expected to solve microscale combustion in the near future. The applications of optical measurements to micro/mesoscale combustion are only at the beginning. Temperature measurements of oscillating flames in a heated channel by OH two-line method, which enriched our understandings on the dynamic flames, were performed [102].

Finally, it is emphasized that further fundamental studies on microscale combustion on extremely small scale targets, non-equilibrium effects, wall quenches and their control, in addition to stability boundaries, mild or flameless combustion in collaboration with cutting-edge computations and laser diagnosis are strongly encouraged for flexible, innovative new designs and concepts for novel combustion technologies.

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